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From Shutdowns to Solutions: Optimizing Island Based Well Intervention Operations in High H₂S Environments

D. Londono, A. Otubaga, G. Correia, and M. Bukhari, ADNOC offshore

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Abstract

This paper presents the development of a safer and more efficient methodology to perform slickline interventions in high Hydrogen Sulphide (H₂S), high-pressure wells located on offshore artificial islands in UAE. Initially, at the end of each run into the wellbore, the only available and safe option to dispose of the sour gas present in the pressure control equipment (PCE) was to bleed off to the island's green burner. The use of the island's green burner is an operation that requires specific wind conditions, both in direction and speed to avoid combustion gases reaching back to the island and/or accommodation.

As a result, waiting on weather (WOW) became a frequent occurrence in the day by day. There was an associated and additional safety risk of having the pressurized PCE containing high-H₂S hydrocarbons left suspended from the crane for extended durations compromising permanent presence of personnel on the site for both the crane and the Slick Line crew. This significantly impacted operational efficiency, contributing to Monthly NPT due to WOW to bleed off the PCE during well interventions estimated up to 10% of operating time.

To mitigate these efficiency and safety challenges, a series of strategies were progressively implemented and evaluated across multiple jobs in the islands. Some strategies initially focused on removing the H₂S from the PCE before breaking containment, while others aimed to prevent the presence of H₂S in the PCE throughout the entire operation. This paper explains each strategy in detail, including the implementation process and results, supported by schematics to enhance understanding. The goal is to provide a proven and transferable methodology for similar high-risk environments with similar challenges where this solution could be implemented.

Introduction

Conducting routine slickline, electric line, and coiled tubing interventions on high-H₂S wells – averaging 3% to 5% H₂S with wellhead pressures (WHP) ranging from 1,500 to 3,800 psi – on SARB field offshore artificial islands housing over 280 personnel per island presents significant safety and operational challenges. In a Quantitative Risk Assessment (QRA) carried out by Bureau Veritas in 2021, the SARB artificial islands were classified as Amber zone, signifying the risk associated with gas release from the wellpad area (Sambhar et al., 2021).

Even basic tasks, such as bleeding off pressure from the pressure control equipment (PCE) at the end of each run, were complicated by the facility's original design, which included only a single burner for safe sour gas release and disposal. This constraint required favorable wind conditions – both in direction and speed – to safely route and burn gas to the burner in such a way that the combustion gases created were directed away from the island. The limitation on waiting on the right weather conditions often leads to operational shutdowns lasting from only a few minutes to several days in worst cases making impossible to have continuous operating conditions.

Cold venting sour gas – bleeding off the gas with H₂S directly to the atmosphere at the wellsite – was strictly prohibited in the islands due to stringent safety regulations developed, first and most importantly, to protect personnel performing the job and second, to avoid triggering the island emergency shutdown system which uses multiples H₂S and LEL sensors located around all Production Trees and Production/Injection manifolds in the island. Such a shutdown would result in the closure of all producing and injector wells on the island, leading to major production losses with significant financial implications.

After identifying the significant time and cost impacts on well intervention jobs caused by these operation shutdowns caused by weather limitations, the construction of a second burner was proposed and recommended as the easiest solution. If present, a second burner will allow routing and burning the gas in a safe way ensuring that the combustion gases are not going to impact the personnel onboard under any weather conditions (common practice during high H₂S well testing interventions in onshore environments). However, the high cost of the project estimated in around US 60 million led to its immediate cancellation, leaving no option at that time but to continue relying on the single available burner.

Following specialized software simulations, burner use was restricted to conditions with a minimum wind speed of 5 knots to ensure safe H₂S dispersion in the airflow, along with wind directions that would carry combustion gases away from the island to avoid personnel exposure. As shown in the figure below, when the wind direction is from the green-shaded directions, it was safe to operate the burner. However, when the wind direction is from the red-shaded directions, burner use was prohibited, as combustion gases including the toxic sulfur dioxide gas (SO₂) – and potentially the deadly hydrogen sulfide gas (H₂S) itself released in the event the flare goes off – could drift back toward the island's field, accommodation and office areas with the potential to reach an important number of persons. The yellow-shaded zones represented transitional areas, where burner use might be permitted occasionally, depending on the precise wind direction and speed.

Although the islands were designed and constructed based on the predominant wind direction of the region – making it common to have wind from the right direction – wind speeds were often below the required 5 knots, making it impossible to use the island burner. This environmental dependency significantly impeded operational efficiency and highlighted the urgent need for alternative gas-handling strategies during regular well interventions performed in the islands.

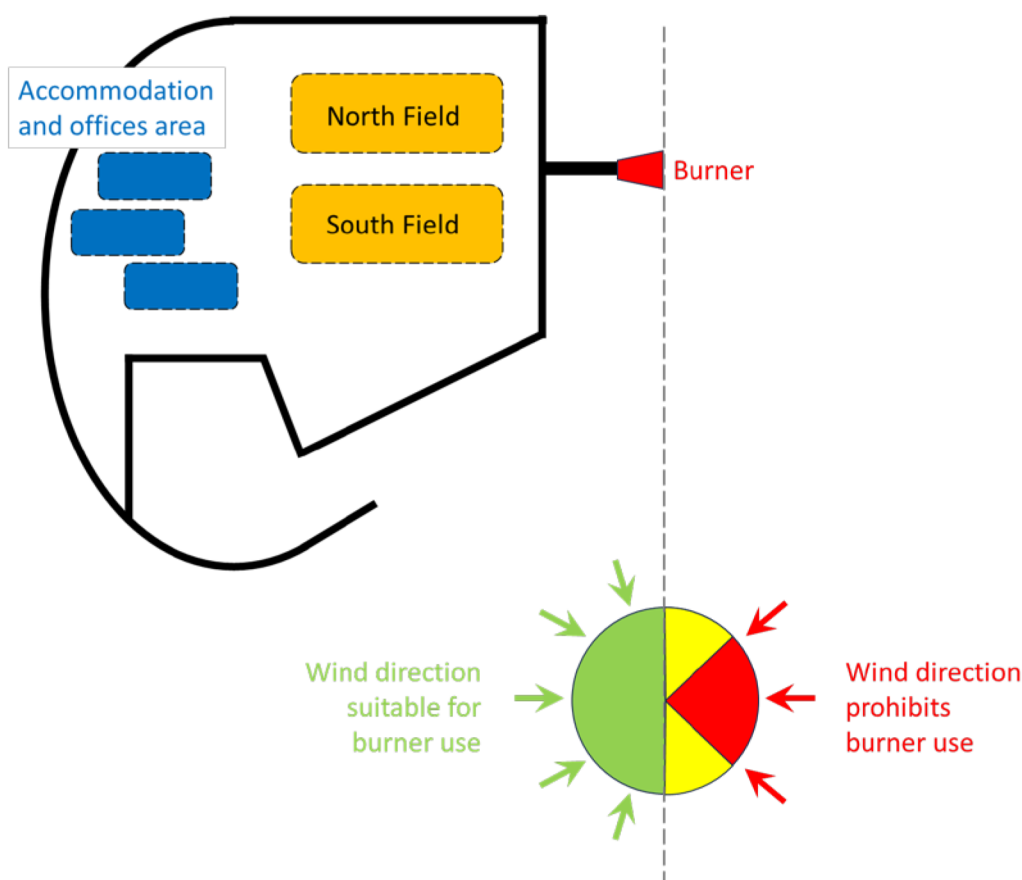


Figure 1—Artificial island layout and green vs red wind directions

The situation was especially critical during slickline interventions, being the most frequent and typically the fastest type of intervention. As an example, a simple slickline drift run to the SCSSSV depth at 300 ft which normally takes less than 30 minutes, can potentially result in significant non-productive time (NPT) of several hours, or even days in the worst cases, while waiting on weather to safely break the lubricator connections and lay down the toolstring at the end of the job or in preparation for the next run. Finding the right operational solution for this limitations were not optional.

Initial Strategy

At the end of each slickline run on a producer well, once the tool string was secured inside the lubricator, the production-tree flow-wing valves and the production choke were opened to bleed the pressure in the pressure-control equipment (PCE) from roughly 2,000 psi (average WHSIP) down to the production-header pressure of about 600 psi. Further reduction was impossible because the production-header pressure could not be lowered any further.

With approximately 600 psi of sour gas still trapped in the lubricator, the first mitigation strategy aimed at reducing non-productive time (NPT) and associated safety risks, was to scavenge the H_2S . To achieve this, water treated with an H_2S scavenger was pumped into the PCE and pressurizing the system to around 4,000 psi. After allowing sufficient time for the scavenger to react, the pressure was bled back down to the production header pressure (approximately 600 psi).

This approach proved largely ineffective. The water-scavenger mixture merely pushed the gas cap upward inside the lubricator, providing little contact between the scavenger and the H_2S -laden gas. Several displacement cycles were often required, and even then, the H_2S could not be reliably removed in most of the cases.

The method was not feasible on gas-injection wells, where the PCE pressure at the end of the run (+/- 3,500 psi) cannot be reduced to the production header because the well is tied into the high-pressure gas-injection header. Since most than half of the slickline jobs are performed on these gas injector wells, the negative impact on operational efficiency remained significant.

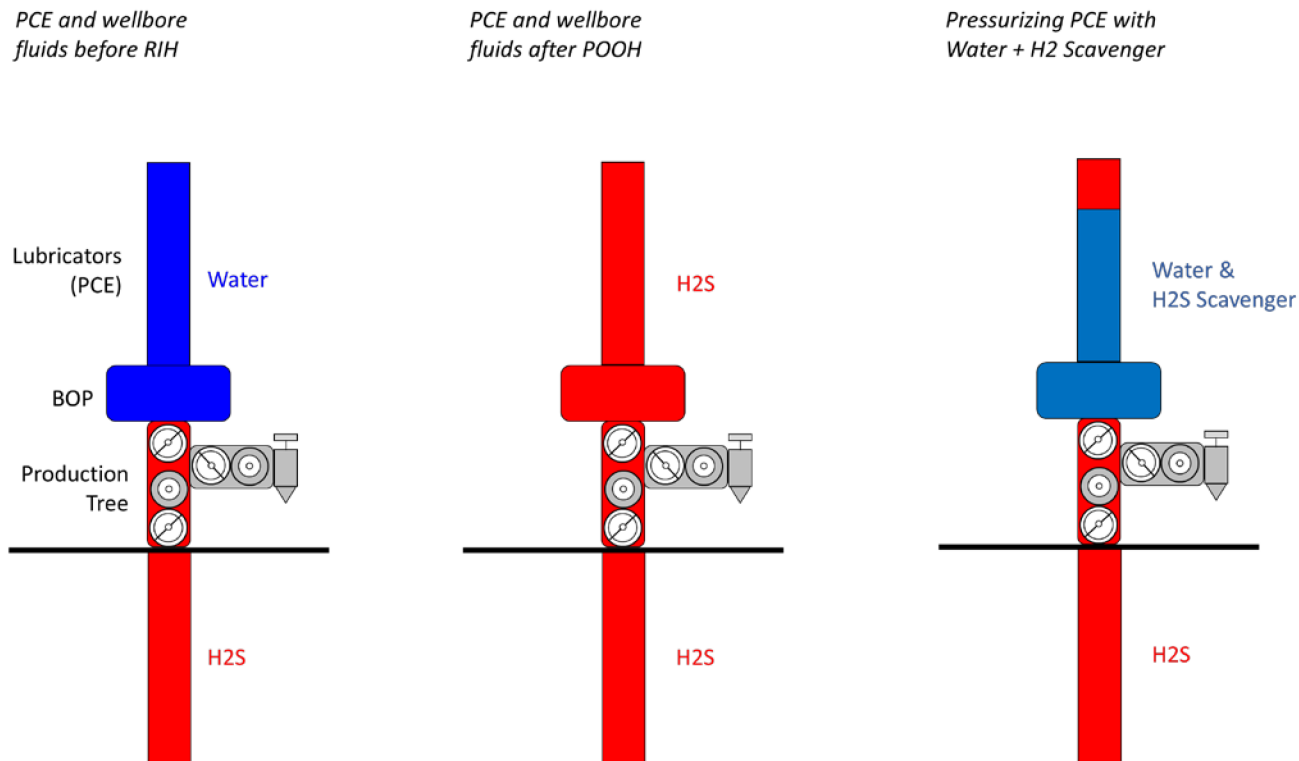


Figure 2—Initial strategy: Pressurizing PCE with water + H₂S Scavenger

Second Strategy

In some jobs performed in producer wells, it was possible to rig up surface lines and connect the liquid nitrogen (N₂) pump permanently present in the island to the slickline PCE. At the end of each run in the wellbore and to be able to break the lubricator, the strategy consisted of pressurizing the lubricator with N₂ and allowing time for the H₂S – the heavier gas – to settle at the bottom of the PCE. After some time, when the pressure was released to the production header, the H₂S would be expelled from the lower part of the PCE, leaving nitrogen – being lighter – at the top of the lubricator.

This process was typically repeated three times, with settling periods of 10 minutes, 10 minutes, and 1 hour, respectively, until only N₂ remained in the lubricator. At the end, the nitrogen present in the lubricator was vented to the atmosphere while continuously monitoring for the presence of H₂S. If H₂S was still detected, the process was repeated till ensuring no additional H₂S was present in the PCE.

Although this strategy is technically supported for the differences in gas densities as described, in practice there were multiple cases where it was not effective because of the small but continuous pass of the H₂S-rich gas above the Production Tree valves. If this last occurrence is not present, meaning the Production tree valves are leak-tight, this strategy proved fully effective for producer wells; however, same as the initial strategy using water & H₂S scavenger, it could not be applied to gas injector wells due to the inability to bleed off to the production header. As a result, overall efficiency remained still negatively impacted after its implementation.

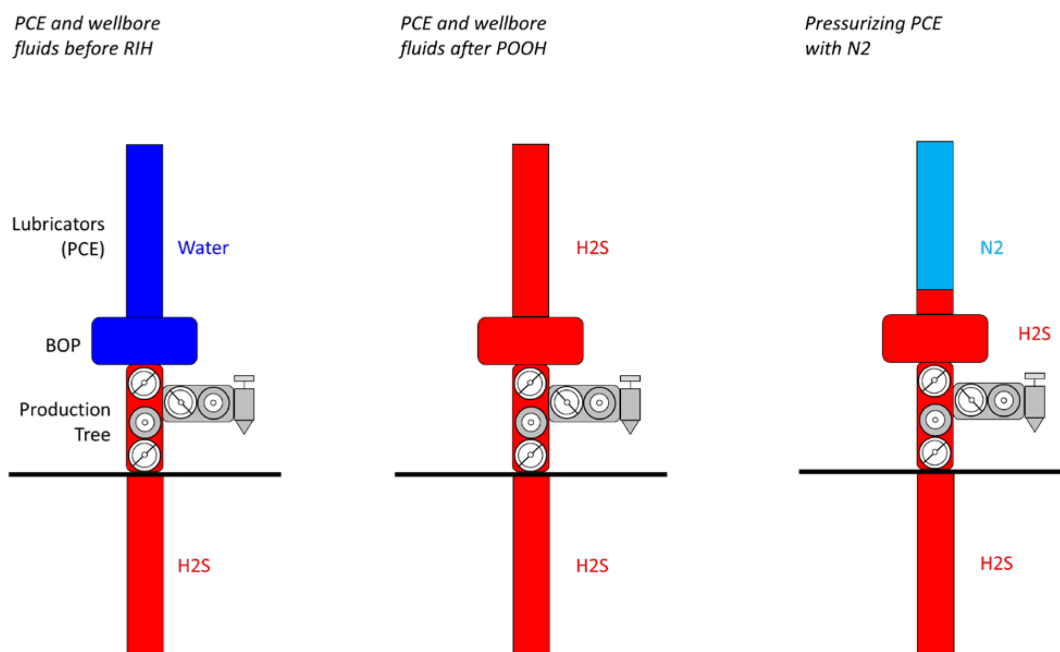


Figure 3—Second Strategy: Presurizing PCE with N₂

Third Strategy

The third mitigation approach was intended to push down the wellbore the H₂S-rich gas initially present below the Production tree valves. To accomplish that, a 5-barrel slug of water & H₂S scavenger followed by 5 barrels of pure water were bullheaded into the wellbore before each run – immediately after opening the HMV, once the PCE pressure test with water has been successfully completed.

This strategy aimed to prevent the upward migration of H₂S gas into the slickline lubricator during the job execution and proved to be highly successful but was limited to shallow slickline runs targeting the surface-controlled subsurface safety valve (SCSSV) usually present in the completion at around 350 ft from surface. In cases where the SCSSV was at around 1,000 ft from surface, this strategy was also successful if no important delays were observed during the job execution.

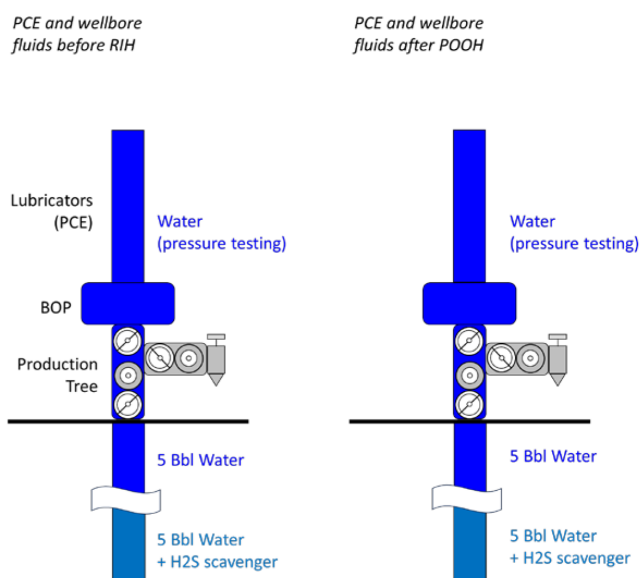


Figure 4—Third Strategy: Bullheading H₂S scavenger + water slugs into wellbore (shallow runs)

However during deeper interventions – such as production tubing drift runs or downhole gauges setting/retrieving which are also very common interventions in the field – the high H₂S gas had enough time to migrate back to surface through the scavenger and water slugs during the time required to perform the job.

Once the toolstring was on the surface, the PCE was full of gas containing H₂S making it necessary to pressure up the PCE with N₂ to get rid off the H₂S as shown in figure 5. Although some efficiencies were obtained when working on shallow depths, the method's effectiveness showed no benefits during deeper interventions, so its overall success was still limited making it necessary to find a better strategy to minimize the operation shutdowns.

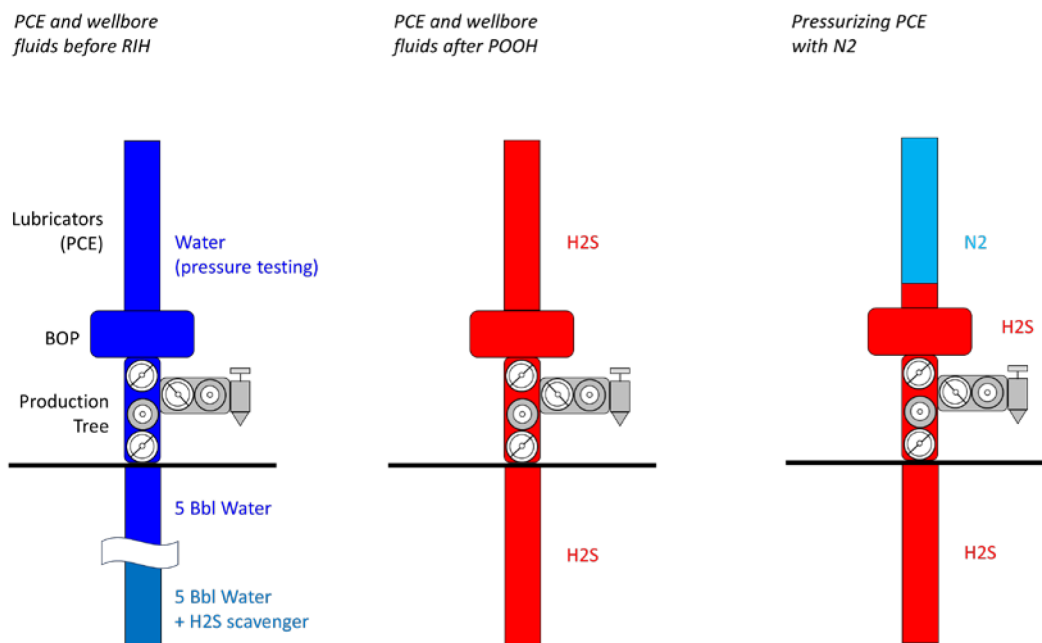


Figure 5—Third Strategy: Bullheading H₂S scavenger + water slugs into wellbore (deep runs)

Fourth and final strategy: N₂ cushion in the wellbore

A major breakthrough in enhancing both safety and operational efficiency was achieved with the implementation of a nitrogen (N₂) gas slug, operationally named as N₂ cushion, pumped into the wellbore immediately before running in hole after completing the PCE pressure test with water.

In the practice, the operational procedure performed during the jobs is:

- Once the pressure test with water is completed, the N₂ pump is cooled down and engaged
- Having only the Hydraulic Master Valve closed, the line was pressurized to around 200 psi above the Wellhead shut in pressure which is the pressure below it
- The HMV is opened and once it is observed a pressure drop for such 200 psi added on purpose above the valve, the N₂ pump started pumping N₂ on lower rates (initial pumping rate around 300 scfm)
- Once the HMV is confirmed visually to be fully open, the rate is increased to around 1,000 scfm

N₂ volume pumped was initially 150 Gal of liquid N₂ which depending on the WHP and the well production tubing size could be equivalent to fill around 1,000 ft in the wellbore with the N₂ cushion. After few jobs of this strategy implementation, it was observed necessary to pump 50 Gal more for each additional run to be performed in the job to ensure a N₂ cushion permanently present in the wellbore long enough to avoid any H₂S reaching surface when pulling out of hole the toolstring through it. This is because there is a sort

of mixing or dynamic contamination at the interface between the H₂S-rich gas below and the N₂ gas above it in the wellbore caused by the passage of the toolstring on the way out.

Based on the first jobs, the lessons learned regarding the total volume required was that for jobs with only one run planned, 150 Gal were pumped; for jobs with two runs planned, 200 Gal were pumped and for jobs with three runs planned (or more), 250 Gal were pumped, being this the maximum volume used so far.

Acting as a cushion, the N₂ displaced the sour gas downward inside the wellbore and remained at the top due to its lower density. During the job execution, this inert gas layer passed above the water used previously for pressure testing effectively isolating the PCE from direct contact with H₂S during the whole job execution.

At the end of each run, the N₂ present in the PCE was safely vented under controlled conditions to a tote tank located at a prudent distance from the Production tree, with no risk to personnel regarding exposure to toxic or deadly gases, eliminating the need for wind-dependent flaring at the island burner and removing all the associated delays and limitations observed in the previous strategies.

A relevant additional advantage of the N₂ cushion implementation from an overall safety perspective is that any leak in PCE during the operations would initially release inert nitrogen, allowing early detection and rapid response and containment before H₂S could pose a risk to personnel in the surrounding area and/or in the island itself.

Since its implementation in August 2024 till June 2025, this technique has been applied in close to 80 Slickline well interventions across both islands with excellent results: ZERO cases of any residual H₂S presence detected in the PCE while the N₂ cushion is being released to the atmosphere at the end of each run.

This strategy is applicable to both oil producers and gas injector wells, making it an ideal solution for all slickline jobs, and even for certain non-flowing electric line (E-line) applications, such as Multi Finger Caliper (MFC) logs performed in the well. Although it cannot be applied to jobs where the well must be flown as most of E-line and coiled tubing interventions, its applicability on slickline jobs certainly impacted the overall well interventions efficiency given the high volume of slickline interventions relative to the other intervention types.

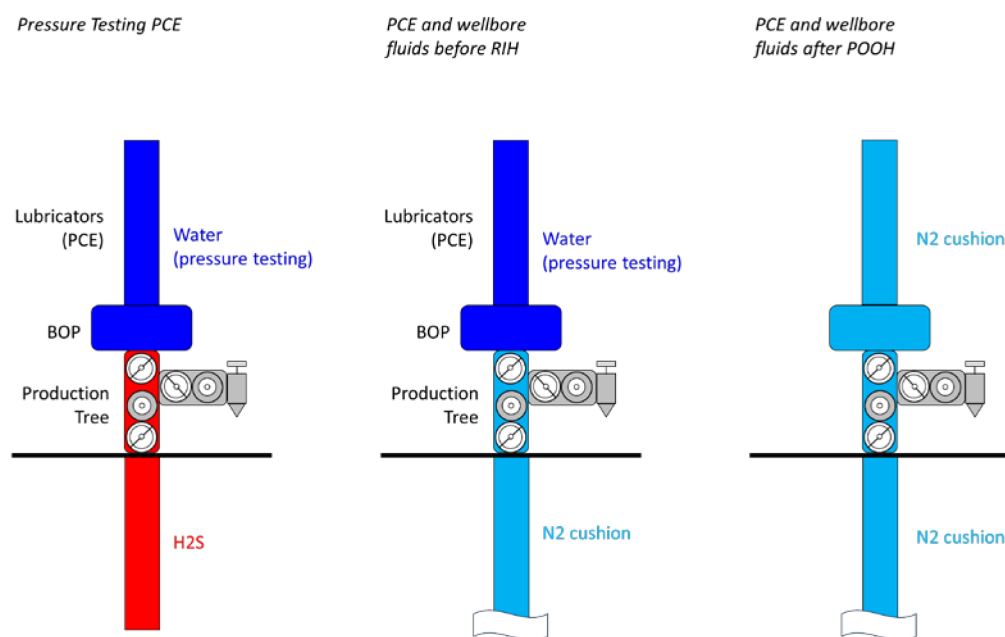


Figure 5a—N₂ cushion implementation

Weather NPT and Cost Implications

Below plots indicates the NPT associated waiting on favourable weather conditions to bleed off slickline lubricator pressure on two Island operation.

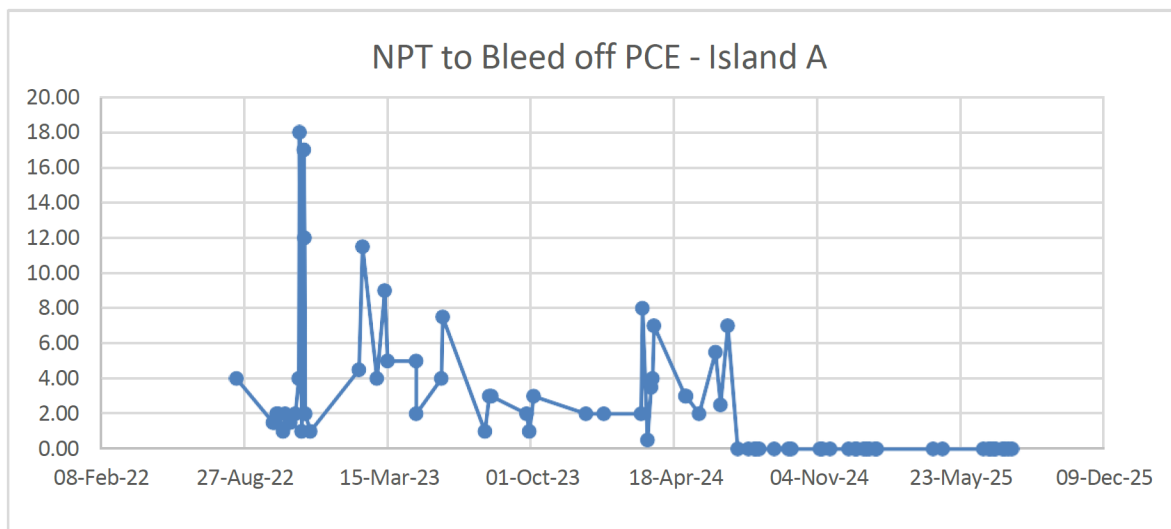


Figure 6—ISLAND A – Total NPT Attributed to Waiting on Weather to Bleed off PCE Pressure

As shown in the above plot, significant NPT was observed during the period Aug 2022 to Aug 2024. However, since the implementation of the N₂ purging system the weather related NPT to bleed off lubricator pressure eliminated for Island A Slickline operation.

A similar trend in the NPT was observed for Island B operation as shown in Figure 7. The impact for Island B was significant since Island B slickline activity was almost double of Island A. N₂ purging solution was implemented for Island B around the same time as Island A (Aug 2024) which resulted in complete elimination of NPT barring one operation in Aug 24 where 3 hours of NPT was reported due to multiple runs and team ran out of N₂ volume during last run.

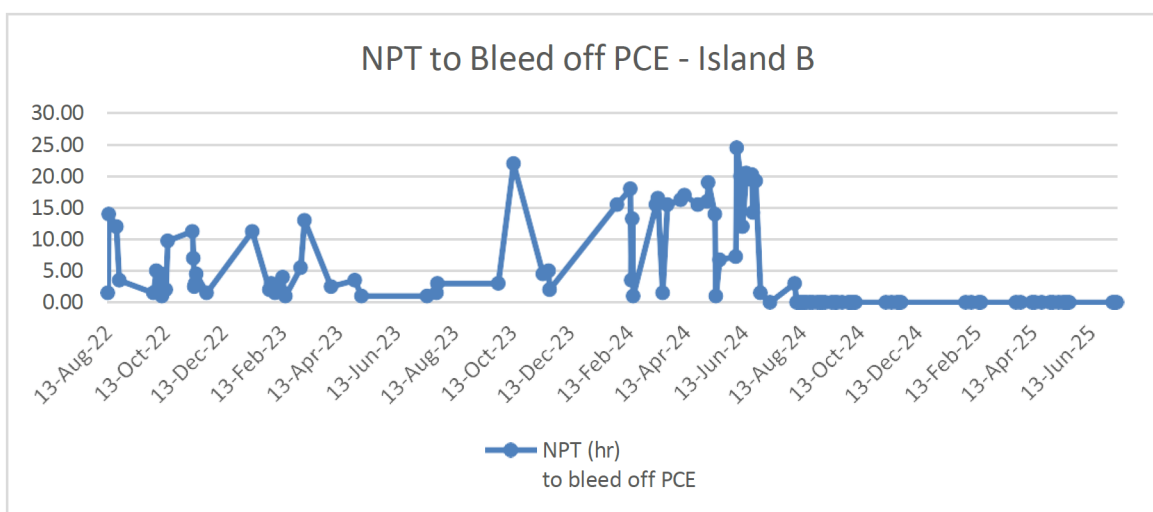


Figure 7—ISLAND B – Total NPT Attributed to Waiting on Weather to Bleed off PCE Pressure

Figure 8 shows Cumulative NPT reported for bleed off operation and # of Slickline operations per Island during Aug 22- Aug

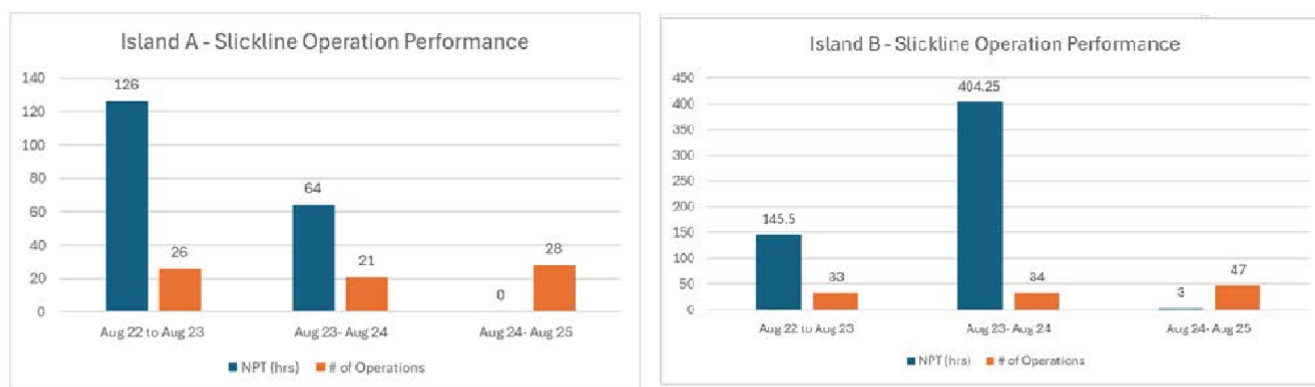


Figure 8—Cumulative NPT and Slickline Operations per Year

NPT levels for Island B increased substantially during Aug23-Aug24 period which triggered the team to find solutions to avoid weather related NPT and streamline slickline operation. The result was N2 purging system which prevented flaring of lubricator gas cap and weather impact was eliminated. Aug 24 – Aug 25 cumulative NPT record clearly shows that although the operation frequency is same, but the NPT was eliminated completely.

Conclusions

- The implementation of the Nitrogen cushion during slickline interventions on high H₂S wells in single burner offshore artificial islands in UAE, improved significantly the overall operational performance by completely eliminating the NPT caused by waiting on weather to safely bleed off the lubricator pressure at the end of each run.
- The implementation of the Nitrogen cushion also eliminated gas flaring to dispose of H₂S gases contained in the PCE at the end of each run for the applicable slickline & E-line operations. This is in line with one of ADNOC's strategic priorities of driving decarbonization, minimizing carbon footprint to achieve 25% greenhouse gas emission reduction by 2030.
- The Nitrogen cushion technique can be applicable to any high-risk hydrocarbon field or facility where gas release poses a threat to personnel. The proactive approach by eliminating sour gas presence at surface is evident to turn a hazardous constraint into a manageable, repeatable process.
- The incremental increase in operational efficiency and resources utilization was successfully achieved by targeting the elimination of weather-related NPT for bleed off activities.

Nomenclature

H ₂ S	Hydrogen Sulphide
HMV	Hydraulic Master Valve
LEL	Lower Explosion Limit
MFC	Multi Finger Caliper
NPT	Non Productive Time
PCE	Pressure Control Equipment
POOH	Pull Out Of Hole
QRA	Quantitative Risk Assessment
RIH	Run In Hole
SCSSSV	Surface Controlled Sub Surface Safety Valve
WHP	Wellhead Pressure
WOW	Wait on Weather

References

Sambhar D., Shekhar S., (2021) *Quatitative Risk Assessemnt (QRA) for Al Qatia & Bu Sikeen, Bureau Veritas – QRA Report.*